Scrum Agile Methodology for Final Year Projects of School of Systems and Technology

School of Systems and Technology

University of Management and Technology

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Final Year Project teams at School of Systems and Technology starts in the seventh semester. Students form groups and submit project proposals. Each group works under the supervision of an assigned faculty member called Final Year Project Supervisor. Once Project proposal is approved, Project teams use scrum methodology to carry out their project.

Roles in Scrum Methodology

There are three roles in Scrum methodology.

1. **Product Owner:** This role is assigned to FYP Supervisor. This is the person Who finalize the requirements. This person is the representative of your customer or customer, if you are developing a project for some real customer.
2. **Development team:** The team of students Who design, develop and tests the application.
3. **Scrum Master:** This is the person among Development team team who is responsible for following Scrum methodology in team. At the start of each sprint, Development Team would choose a Scrum Master among themselves.

Defining Product Backlog

1. The Scrum Process starts with Product Owner (Project Supervisor) and Development Team(Student Project Team) conduct a series of meeting in order to create a Product Backlog document. (Its format is given in the FYP Documentation Template Version 1.0) and Figma UI/UX Designs related to user stories mentioned in the product backlog document.
2. Product Owner and whole development team must be present in those meetings to finalize the requirements that are then written in the Product Backlog.
3. Team members and Product Owner need to identify high-level requirements, such as identifying epics and user stories. Product Backlog features and key non-functional requirements are written in the form of user stories.
4. The analysis will not be perfect. The team regularly do Backlog grooming, also known as backlog refinement.
5. The Epic and User stories would be maintained on Jira by the Scrum master of Development team.

Sprint

In Agile project management's scrum methodology, whole project is divided into a series of Sprints. A sprint is a fixed-length period during which a team works to complete a set of tasks and deliver a potentially shippable increment of product. Sprints are typically one to four weeks long, but two to three weeks is a common duration.

Development team and product owner need to decide Sprint duration first that can be 1 - 4 weeks (pick the timebox, and stick to it for all Sprints).

Number of Sprints may vary according to the scope of the Projects and duration of a single Sprint that is decided by the Product Owner (FYP Supervisor) and Development team for the duration of two semesters of FYP-I and FYP-II.

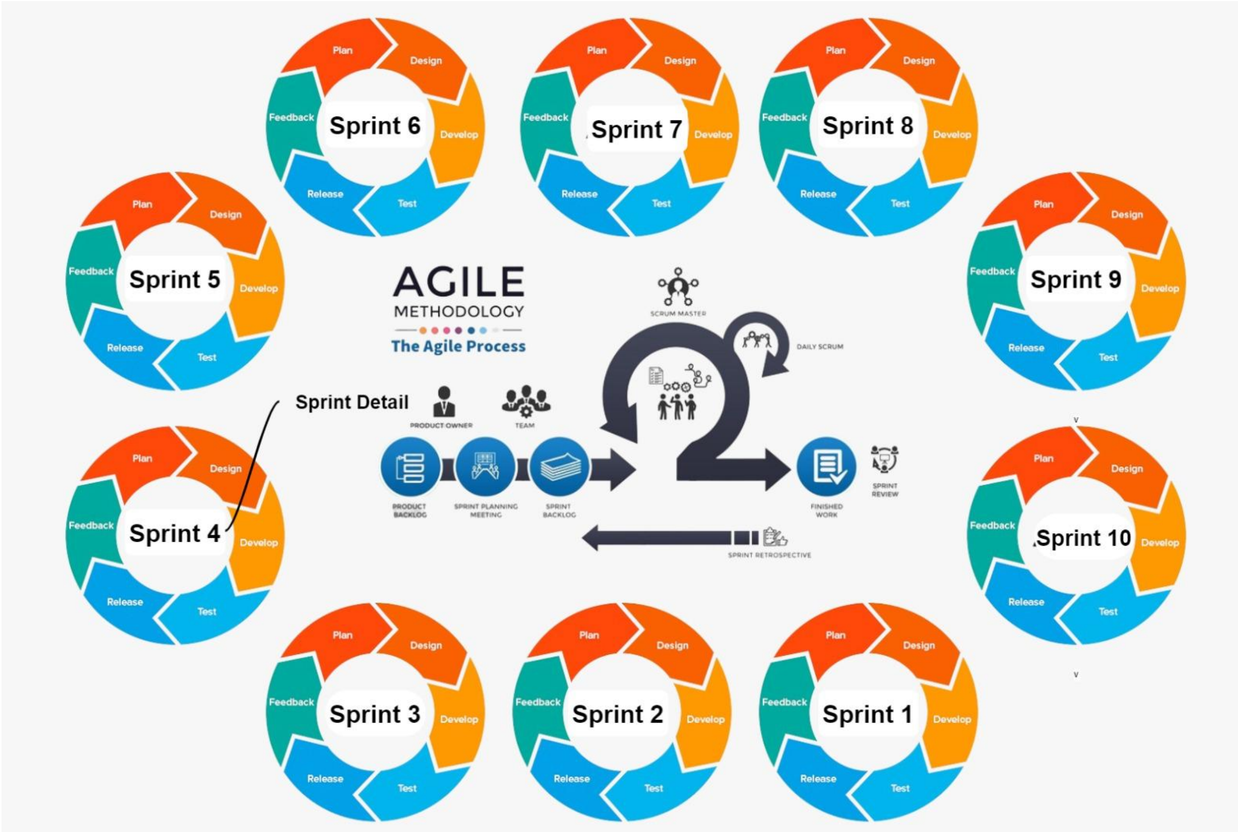

Figure 1: Scrum Process

Meetings, Roles, Document and Sequence of Activities in a Sprint

The following sequence of activities are carried out by Product owner and development team in a single Sprint.

Sprint Planning Meeting

Selection and Role of Scrum Master:

1. At the start of each sprint, Development Team would choose a Scrum Master among themselves.
2. The Scrum Master's responsibility would be to make sure that team follows Scrum methodology and do daily scrum meeting and manage tasks in Jira, selenium test case management and take care of GitHub repository for that Sprint.
3. It must be rotated to each member in each Sprint.

Role of Product Owner and Development Team in a Sprint

1. The product owner is responsible for identifying which features are most important, and the team is responsible for determining what is possible within the sprint.
2. The product owner and development team negotiate which product backlog items the team will attempt this sprint.
3. Development Team asks the Product owner to pick user stories from the Product Backlog that have a blending of these three qualities:

A) Architecturally significant (if implemented, we are forced to design, build, and test the core architecture),
B) high business value (features business really cares about)
C) high risk (such as "be able to handle 500 concurrent transactions")
These user stories are chosen to design, build, and test within a specified time (1-4- week Sprint duration).

4. If it is impossible to complete a certain key item within 1-4 weeks (whatever is Sprint duration), the team works with the product owner to break the feature down into doable pieces.

Sprint Backlog

1. Team should take some time and do intensive detailed analysis of the selected user stories and non- functional requirements from product backlog and breakdown these user stories into tasks that are then used to create Sprint Backlog (Its format is given in the FYP Documentation Template Version 1.0)
2. Once the meeting has finished the goals for the Sprint cannot be changed.

Sprint Planning Meeting minutes

Development team must write each Sprint Planning Meeting minutes (Its format is given in the FYP Documentation Template Version 1.0) and would be a part of each group’s Final documentation.

Sprint Activities

Design

1. On the first two days, development team put on design hat and do modeling and design work as a team or in pairs, sketching UML class and sequence diagrams at whiteboards and/or papers and reviewed by Product owner.
2. For each Sprint Development team must include a Design class diagram, multiple Sequence diagram as part of their design (Its format is given in the FYP Documentation Template Version 1.0) and would be a part of each group’s Final documentation.

Development

1. Then the developers take off their "modeling hats" and put on their "programming hats." They start programming, testing, and integrating their work continuously over the remaining weeks, using the UML diagrams as a starting point of inspiration, knowing that the models are partial and often vague.
2. These are refined as they code.
3. The actual design lies in the final code.

Testing

1. Decision tables are made for complex scenarios. Much testing occurs: unit, acceptance, load, usability, and so forth.
2. For each Sprint Development team must include Test cases and Decision tables (if needed) (Its format is given in the FYP Documentation Template Version 1.0) and would be a part of each group’s Final documentation.

Daily Scrum Meeting

The heartbeat of an Agile project is the daily meetings which are commonly referred to as the "Scrum." Each work day at the same time and place, team members stand in a circle and take turns answering the following key questions:

1. What have you done since the last Scrum?
2. What will you do between now and the next Scrum?
3. What is getting in the way (blocks) you from performing your work as effectively as possible?

For each Sprint Development team must maintain a log of the Daily Scrum meetings in the format given separately and viewed by Product Owner at the end of Sprint and submitted to project office for Scrum compliance by group. This is not part of your FYP documentation template.

Descope (if needed)

In the middle of the Sprint, the team evaluates, if the original Sprint goals can be met; if not, de-scope the iteration, putting secondary goals back on the "to do" list or Product Backlog.

Conclude Sprint:

1. On Wednesday of the last week there's a code freeze; all code must be checked in to GitHub repositories, integrated, and tested to create the iteration baseline.
2. On last week's Thursday morning, demo the partial system to product owner and external stakeholders (if any), to show early visible progress.
3. Feedback is requested.

Sprint Review and Sprint Retrospective Meetings

1. On last week's Friday Morning, Sprint Review and Sprint Retrospective meetings are held.
2. For each Sprint Development team must write each Sprint Review and Sprint Retrospective minutes in the format given in the FYP documentation template and included in their final documentation.